ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE H

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Transistors, analogue and digital signals, radio control, stepper and AC/DC motors, a forward-reverse relay circuit, hydraulic definitions and symbols, regenerative circuits, a two-hand electro-hydraulic press control, and lock-out and daily checks.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw all circuits and symbols neatly, using a ruler and standard symbols.
5. Write neatly and legibly.

QUESTION 1

1.1 Define the term transistor and name its TWO main uses. (4)

1.2 Explain the difference between an analogue signal and a digital signal. Draw a simple sketch of each. (6)

1.3 Draw a labelled circuit diagram of an NPN transistor used to switch a relay. Include the base resistor and a protection diode across the relay coil. (6)

1.4 Give TWO advantages of using a microcontroller instead of hard-wired relay logic to control a machine. (4)

[20]

QUESTION 2

2.1 Explain how radio control works, referring to the transmitter, the receiver and the antennas. (4)

2.2 Explain how a stepper motor turns in steps. (4)

2.3 Give TWO differences between AC motors and DC motors. (4)

2.4 Design a relay control circuit for the forward and reverse running of a DC motor. Use relay K1 for forward and relay K2 for reverse, a START-FORWARD and a START-REVERSE button, one common STOP button, holding contacts and electrical interlocking so that K1 and K2 can never be on together. (8)

[20]

QUESTION 3

3.1 Define the following terms:

3.1.1 Density (1)

3.1.2 Additive (in hydraulic oil) (1)

3.1.3 Cavitation (1)

3.2 Draw the symbol of EACH of the following:

3.2.1 Check (non-return) valve (2)

3.2.2 4/2-way directional control valve, solenoid operated with spring return (4)

3.2.3 Filter (2)

3.3 Explain the difference between an open-centre and a closed-centre 4/3-way valve. (4)

3.4 Explain how a regenerative circuit makes a double-acting cylinder extend faster. (5)

[20]

QUESTION 4

4.1 Explain what is meant by feedback in an electro-hydraulic system. (2)

4.2 Explain the purpose of a pressure switch in an electro-hydraulic circuit. (2)

4.3 A hydraulic press cylinder is controlled by a 4/2-way valve with a single solenoid (Y1) and a spring return. Design the electrical control circuit so that: the press only extends while BOTH two-hand buttons S1 and S2 are held in; it retracts as soon as either button is released; it can only extend when the safety guard is closed (guard switch S3 closes when the guard is shut); an emergency stop removes the power; and a red lamp H1 lights up while the guard is open. (10)

4.4 Give FOUR advantages of a PLC over hard-wired relay control. (4)

4.5 What does an electro-hydraulic servo valve do? (2)

[20]

QUESTION 5

5.1 Explain what is meant by a systematic approach to fault-finding. (2)

5.2 Explain how a pressure gauge can be used to help find a fault in a hydraulic system. (4)

5.3 An electric motor does not start when the START button is pressed. List, in a logical order, FOUR checks you would make. (4)

5.4 Explain the lock-out and tag-out procedure. (5)

5.5 List FIVE daily checks that an operator should carry out on a hydraulic machine before starting it. (5)

[20]

TOTAL: 100

END OF QUESTION PAPER